

# A versioned, leakage-audited USGS *Did You Feel It?* benchmark for retrospective severe felt-intensity prediction

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## Abstract

Reliable comparison of models that estimate earthquake felt intensity requires a stable reference dataset and evaluation rules that prevent related events from crossing data splits. We introduce version 1 of a single-source USGS “Did You Feel It?” (DYFI) benchmark for retrospective prediction of severe felt intensity (event-maximum community decimal intensity  $\geq 6$ ) from earthquake source metadata. The resource combines a frozen dataset, a six-category leakage audit, prespecified space-time sequence grouping, outcome-independent analysis roles, endpoint sensitivity analysis, fixed baselines, calibration-aware scoring, sequence-cluster bootstrap uncertainty, and a protected temporal evaluation. The frozen  $M \geq 5$  snapshot contains 7,520 records, 2,362 eligible events, and a 430-event temporal holdout. For the fixed random-forest diagnostic, the observed random-versus-sequence-grouped differences were small and slightly favored sequence grouping (Brier-score difference  $-0.006$ ; area under the receiver-operating-characteristic curve [AUROC] difference  $-0.012$ ); we report this transparently as a measured null. Every learned baseline improved on the no-skill reference, while the direct comparison between the two lowest-Brier models remained unresolved. The benchmark provides a versioned, leakage-audited reference for future methods, and every reported value can be reconstructed exactly from the frozen analysis files.

**Keywords:** Did You Feel It?, macroseismic intensity, benchmark resource, leakage-safe evaluation, sequence-grouped cross-validation, calibration, reproducibility

## 1 Introduction

The U.S. Geological Survey “Did You Feel It?” (DYFI) system converts crowd-sourced felt reports into a numeric community-decimal intensity (CDI) for each earthquake, providing a dense, decades-long record of observed shaking that complements sparse instrumental networks [1, 13, 20, 21]. Because the felt-intensity outcome of an event is only realised after responses accumulate, a natural retrospective question is how well the *severe*

outcome of an event—an event-maximum intensity in the damaging range ( $\text{maxCDI} \geq 6$ )—can be anticipated from outcome-blind source metadata that are known at or near origin time (magnitude, depth, location, origin time). To protect the temporal holdout, however, the primary models use only magnitude and depth; location and origin time appear only in restricted sensitivity analyses. This is a descriptive-retrospective estimand, useful for characterising how much of the severe-shaking signal is carried by catalogue

parameters alone, and it is distinct from any real-time, operational, or causal claim.

A recurring methodological problem undermines the machine-learning literature on this and adjacent tasks. Models are trained and evaluated on datasets assembled ad hoc from the earthquake catalogue and split either at the record level or by a single random train/test partition. Random splitting ignores the dependence structure of seismicity: aftershocks of a mainshock share source parameters and overlapping felt fields, so a random split can place near-duplicate events on both sides of the partition and inflate apparent skill—a textbook form of data leakage [8] that grouped or blocked cross-validation is designed to remove [14]. Recent macroseismic-intensity studies illustrate the pattern: models are fit on single-system instrumental or DYFI data under random or stratified partitions, with no released, leakage-audited, event-grouped evaluation resource that a later user could reuse without reconstructing a contaminated split [10, 11]. The consequence is that reported numbers are hard to reproduce and easy to over-read.

## 1.1 Contributions

We address the reproducibility and leakage gap by introducing a versioned benchmark resource centered on a reusable, leakage-safe evaluation contract. Concretely, the resource provides:

1. **A single-source, version-frozen DYFI benchmark** for the exact task “outcome-blind source metadata  $\rightarrow$  later event-maximum  $\text{maxCDI} \geq 6$ ”, frozen to one dated access instant so that any later catalogue revision defines a new version rather than a silent overwrite (Section 3).
2. **An executable six-category leakage audit** that rejects direct-label, post-outcome-ascertainment, identifier-proxy, and downstream-product features, and rejects cross-role sequence straddles and temporal-cutoff violations, accepting *only* the exact source-only feature set; it is exercised by 110 unit tests and rejects every deliberately contaminated test case (Section 4.1).
3. **Prespecified, outcome-independent roles** built on order-stable space-time sequence grouping, with a temporal external holdout

and a secondary geographic check, so that split membership never uses the label (Section 4.2).

4. **A fixed, non-selective baseline suite** with calibration-aware proper scoring, sequence-cluster bootstrap uncertainty, explicit handling of ties and unavailable calibration estimates, and a single-use protected evaluation (Sections 4.4–4.5).
5. **A quantified leakage-inflation diagnostic** (naive random versus sequence-grouped splitting) reported as a descriptive point estimate, together with a fully reproducible release bundle and machine-readable verification record (Section 5).

The learned models establish reference performance for the benchmark. Their comparisons do not identify a winner, and the study makes no cross-system, real-time, lead-time, operational, or causal claim.

## 1.2 Significance

The study is significant because the macroseismic-intensity machine-learning literature currently lacks a released, versioned, leakage-audited evaluation resource for the felt-intensity task. It supplies a reusable evaluation foundation: a version-frozen corpus, an executable leakage audit, outcome-independent grouped roles, calibrated baselines reported without overclaiming, and an exact-reproduction guarantee. Filling this gap lets subsequent work evaluate against a fixed leakage-safe contract instead of re-inventing a contaminated split, which is the precondition for comparable, reproducible progress on the problem. Releasing an evaluation resource of this kind is an established contribution type in applied-science venues: for instance, Urs et al. [15] independently benchmark existing prediction tools against gold-standard datasets and report the gap between claimed and independently measured performance, an evaluation-resource contribution analogous in kind, though in a different domain, to the one made here. This paragraph concerns the study’s significance and the literature gap, not its findings, which are reported separately.

## 2 Related work and novelty

### *DYFI and macroseismic intensity.*

The DYFI system and its CDI construction, outlier handling, and post-event revision behaviour are documented in the foundational and review literature [1, 13, 20, 21]; these establish that CDI is an after-the-event, response-driven quantity subject to ascertainment effects, which is precisely why our protocol uses response count only to determine eligibility and forbids it as a predictive feature (Section 3). We reuse these provenance facts rather than re-deriving them.

### *Machine learning on felt/macroscopic intensity.*

Recent studies predict macroseismic intensity with random forests and gradient boosting from instrumental or catalogue features [10, 11]. Each is a single-system model application evaluated under a random or stratified partition and does not release a reusable, event-grouped, leakage-audited benchmark; Mori and Naso [10] moreover predict from instrumental ground-motion parameters rather than outcome-blind catalogue metadata, and Patelli et al. [11] evaluate a single-system train/test split on seismometric inputs. Their concrete drawback for our purpose is the evaluation contract: a random or stratified split does not remove within-sequence near-duplicate contamination, and none ships an executable leakage audit or a version-frozen resource that a third party can rerun. Our protocol addresses that specific drawback with prespecified sequence-grouped roles and an executable six-category audit.

### *DYFI-based intensity models and concurrent leakage-aware work.*

Closest to our exact data source, Karthik Kumar et al. [7] predict DYFI community intensity from the same catalogue metadata (magnitude, depth, location) with a random-forest regressor on Indian events under a single naive train/test split (reported  $R^2 = 0.829$ ); our resource does not fit a competing regressor but instead supplies a versioned, digitally fingerprinted, event- and sequence-grouped benchmark with an executable leakage audit and a protected single-use evaluation, in place of the naive split that setup uses. Concurrently, Li et al. [9] release a

DYFI-grounded benchmark that simulates Modified Mercalli Intensity for two earthquakes (2014 Napa and 2019 Ridgecrest) with large language models and includes an explicit location-identifier leakage test; our resource differs in task and construction—an outcome-blind source-metadata→later-severe binary classification over a decade-scale  $M \geq 5$  corpus with prespecified sequence-grouped roles, rather than multimodal per-zip-code MMI simulation for two events—so the two are complementary rather than duplicative.

### *Cross-validation for dependent data.*

That grouped or blocked cross-validation is the correct evaluation strategy for temporally, spatially, or hierarchically structured data is established practice [8, 14]; we do not claim this as novel.

### *Novelty delta.*

Conceding the above, the delta this resource supplies that prior work does not is the *released, versioned, hashed, single-source DYFI benchmark* itself: an executable six-category audit covering forbidden fields, proxy variables, and split contamination; a prespecified, sequence-grouped role assignment with a temporal external hold-out, a documented severe endpoint with threshold and aggregation sensitivity, a fixed non-selective baseline suite with a no-skill comparator, a calibration/uncertainty/tie contract, an exact-reproduction guarantee, and a quantified naive-versus-grouped leakage-inflation headline. A current-year search located no released resource occupying this exact combination; novelty is stated as a concrete delta to located work, with no first/only superlative asserted.

## 3 Data, provenance, and outcome-boundary disclosure

### *Source and single-instant pinning.*

The corpus is derived entirely from USGS DYFI and its parent ANSS Comprehensive Catalog (ComCat), retrieved through the International Federation of Digital Seismograph Networks (FDSN) event web service [16, 17]. The acquisition envelope is outcome-independent:  $M \geq 5$ ,

product type `dyfi`, event type `earthquake`, start-time 2015-01-01 to endtime 2025-01-01 (the fixed 2015–2024 decade, with the end bound exclusive), ordered by time. The preliminary count request and the final data request use this same envelope. Because the catalogue evolves, the downloaded data were frozen at one access instant (2026-09-01T06:33:30Z) and assigned a SHA-256 digital fingerprint. A preliminary count changed from 7,575 to 7,576 during the acquisition session, documenting one live-catalogue revision. The final downloaded snapshot contained 7,520 DYFI records, and a count repeated at that same instant also returned 7,520. Thus, the one-record change belongs only to the earlier preliminary checks; it is not a discrepancy in the frozen analysis snapshot. The digital fingerprint of the downloaded bytes, rather than a preliminary count, defines this resource version. The frozen snapshot contains 7,520 DYFI records (Table 1, Fig. 1).

#### ***Retrospective eligibility and post-report ascertainment.***

DYFI intensity is an after-the-event quantity whose reliability grows with the number of felt reports [13]. To keep the label meaningful we apply a data-informed eligibility filter of at least ten responses per event. The response count is used *only* for eligibility and is explicitly forbidden as a predictive feature, because it is a post-outcome ascertainment proxy; this forbiddance is enforced by the leakage audit (family F2, Section 4.1). Eligibility yields 2,362 events with severe prevalence 0.298 (703 severe events). Of these, 2,360 events are assigned to the prespecified analysis roles (Section 4.2) and contain 702 severe events; the remaining two are temporal-straddle exclusions, one of which is severe. That single severe straddle event accounts for the difference between the eligible cohort (703 severe) and the analysis-assigned set (702 severe), so eligibility is not the same as analysis-role assignment (Table 1).

#### ***Endpoint.***

The severe label is severe =  $\mathbf{1}[\text{event-maximum maxCDI} \geq 6.0]$  (intensity VI+). The primary threshold is 6.0, with reported sensitivity at  $\{5.5, 6.0, 6.5\}$  and a near-threshold band of  $\pm 0.5$  (Section 5.2).

#### ***Outcome-boundary disclosure and held-out semantics.***

Only corpus-level outcome-blind counts and inherited aggregate feasibility statistics were exposed before construction; no per-event label informed the endpoint, split, or tuning choices, and role membership is assigned by a prespecified, outcome-independent rule that never uses the label. We therefore describe the external holdout as *held out from model development with aggregate feasibility previously known*, and we deliberately do *not* call it “untouched” or “confirmatory.” Two events examined during source verification are conservatively excluded from every analysis role because their aggregate counts had already been seen (Table 1); the complete list of protocol decisions is reported in Section 5.7. No respondent-level personally identifying information is redistributed; only event-level source metadata, the severe label, the eligibility-only response count, role assignment, and per-event provenance hashes are released. The release bundle follows datasheet-for-datasets documentation [5] and findable, accessible, interoperable, and reusable (FAIR) data principles [22].

## **Benchmark design**

### **4.1 Feature sets and the six-category leakage audit**

The primary *source-only* feature set contains magnitude and depth. It is the only feature set evaluated on the transport holdouts. An *expanded-metadata* sensitivity analysis (baseline B3) additionally uses latitude, longitude, and event year, so it is not evaluated where those variables would disclose the holdout axis. The audit accepts only an approved feature set with the expected fields in the expected order; any extra or unknown field is rejected.

The executable audit enforces six leakage families. Field-level: F1 direct label or its constituents; F2 post-outcome ascertainment or revision (including the response count, eligibility-only); F3 identifier or source proxy; F4 downstream-product proxy. Split-level: F5 sequence cross-role straddle; F6 temporal-cutoff violation. An axis-proxy check additionally rejects origin year under a temporal holdout and latitude/longitude/region under

a geographic holdout, enforcing the B3 restriction. This audit is useful independently of model performance: it must reject every deliberately contaminated test case and accept only the approved clean feature set. It is exercised by 110 unit tests across four suites, all passing, and accepted the source-only feature set in the frozen snapshot.

## 4.2 Sequence grouping and outcome-independent roles

Events are grouped into seismic *sequences* by order-stable single-link space-time connected components (100 km, 30 days); the sequence, not the event, is the unit of role assignment, so after-shocks cannot straddle roles. Roles are assigned by a prespecified, outcome-independent rule (fixed assignment key, temporal cutoff year 2023, development fraction 0.2, geographic-holdout fraction 0.2, five cross-validation folds); the rule provably never reads the label. The primary external holdout is events on or after 2023 (temporal transport); a leave-region-out geographic holdout is a secondary transport check. Digital fingerprints of the completed role assignments are published so that third parties can verify them. The partition is exhaustive and disjoint:  $1,573 + 357 + 430 + 2 = 2,362$  events, comprising 1,573 train, 357 development, 430 external temporal holdout, and 2 temporal-straddle exclusions (Table 1, Fig. 1). Construction (train plus development) is 1,930 events across 1,530 sequences; the holdout is 430 events across 354 sequences.

## 4.3 Baseline suite and exclusions

The fixed suite is B0 no-skill (constant prevalence), B1 magnitude-only logistic regression, B2 source-only logistic regression, B3 expanded-metadata logistic regression (used only in compatible sensitivity analyses), B4 random forest (scikit-learn; 12), and B5 XGBoost [3]. Each learned baseline is tuned *only* by the registered deterministic grouped cross-validation on the construction corpus over a fixed grid, named before evaluation, and carried unchanged to the protected holdout with frozen best parameters. A machine-readable fit log enforces a maximum of 120 model fits; the executed run used 91. Two comparators are excluded for prespecified, reproducible reasons recorded in the protocol-decision table

(Section 5.7): B3 is not evaluated on the temporal holdout because its feature set carries origin year, a temporal proxy; and the Atkinson–Wald (2007) DYFI intensity-prediction equation [1] is excluded because it cannot be applied to the source-only feature set and tectonically heterogeneous population without inventing a site-distance covariate, a per-event tectonic-regime assignment, and a probability link.

## 4.4 Metrics, calibration, uncertainty, and ranking

The primary score is the Brier score [2]; secondary scores are log-loss, AUROC, and calibration intercept/slope [18]. Uncertainty is quantified by a sequence-cluster bootstrap ( $n_{\text{boot}} = 2000$ , 95% intervals) that resamples whole sequences rather than events [4], with a minimum-valid-replicate rule. Calibration and feasibility states are *derived*, not hard-coded: a constant-prevalence predictor has no identifiable calibration slope, so the table reports that value as unavailable. Pairwise ranking uses a paired sequence-cluster bootstrap Brier difference with an explicit tie probability; each pair is reported as resolved (95% interval excludes zero) or unresolved, with a tie-aware joint ordering (Section 5.5).

## 4.5 Protected evaluation and reproducibility

The protected temporal holdout is evaluated exactly once under an automated single-use lock; a second evaluation is refused. After evaluation, two clean reconstruction runs reproduced the recorded result fingerprint exactly, and the frozen holdout predictions were recovered bit-for-bit from the released rows and frozen best parameters (reported in Section 5.7), under a pinned scientific-Python environment (NumPy, SciPy, scikit-learn, XGBoost; 3, 6, 12, 19). All fits, exclusions, subsets, cutoffs, and pairwise comparisons are reported regardless of outcome; the only permitted omissions are the prespecified exclusions above. Computation is local-only under a pinned environment.

## 5 Results

All result-bearing values below are generated from the frozen analysis outputs and checked automatically against their source records; no result cell was hand-edited.

### 5.1 Cohort and roles

Table 1 and Fig. 1 give the cohort accounting. Of 7,520 frozen DYFI records, 2,362 events pass the ten-response eligibility filter (severe prevalence 0.298; 703 severe events), partitioned into 1,573 train, 357 development, and a 430-event temporal holdout, with two temporal-straddle exclusions. Thus 2,360 of the 2,362 eligible events receive analysis roles; they contain 702 severe events, and the 703rd severe event is one of the two straddle exclusions. Separately, two records examined during source verification were removed before the eligible cohort was constructed and are not part of the 2,362-event total. Construction severe prevalence is 0.285 and holdout prevalence is 0.351; the higher holdout prevalence reflects the temporal transport shift and is reported, not adjusted.

### 5.2 Difficulty and endpoint sensitivity

Table 2 and the left panel of Fig. 2 show the endpoint is not an artifact of a single threshold: as the severe cut moves from 5.5 to 6.5, holdout prevalence falls smoothly (0.458, 0.351, 0.260) and the source-only logistic-regression Brier score and AUROC move monotonically (Brier 0.240, 0.181, 0.143; AUROC 0.734, 0.786, 0.815). On the construction corpus the near-threshold fraction ( $|\text{maxCDI} - 6| < 0.5$ ) is 0.164 and the no-skill Brier score is 0.204, bounding the difficulty of the primary endpoint.

### 5.3 Observed split-performance differences

The resource’s headline quantity is the optimism of naive random splitting relative to the sequence-grouped protocol, measured for a fixed random-forest diagnostic under three fold-assignment schemes on the construction corpus (Table 3, Fig. 3). On this  $M \geq 5$  one-row-per-event universe the point estimates are small and slightly

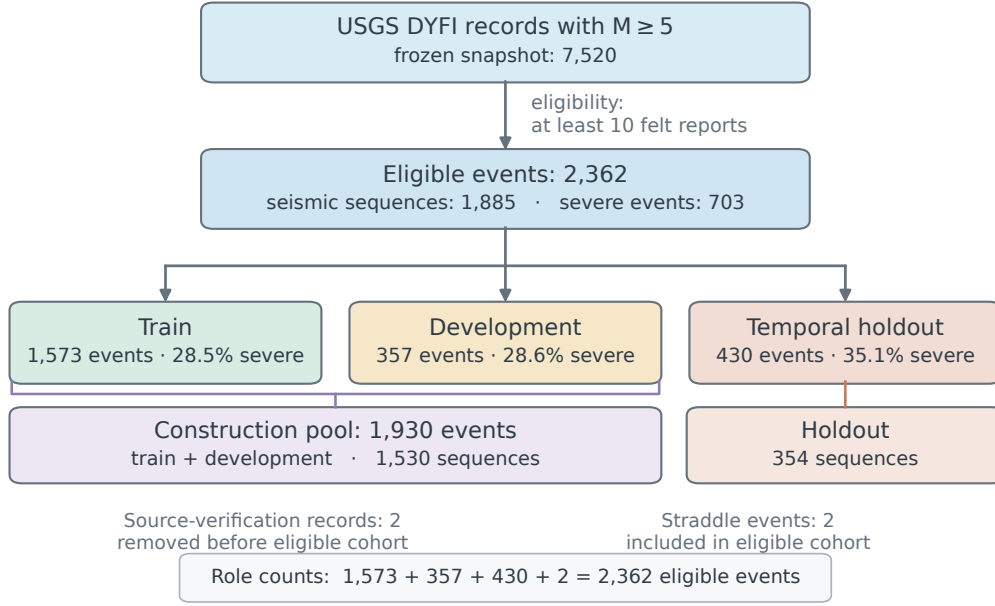
reversed: the sequence-minus-random Brier difference is  $-0.0059$  and the random-minus-sequence AUROC difference is  $-0.0124$ , calculated from unrounded scores. The sequence-grouped split therefore scores marginally *better* than the naive random split, not worse. No uncertainty interval was prespecified for this diagnostic contrast, so we report it descriptively as an honest measured null, not as evidence of equivalence, leakage, or a win. Its explanation is structural: with 1,885 largely singleton sequences (1,530 in construction), there is little within-sequence near-duplicate structure for a random split to leak, so the sequence-grouping choice has little practical effect in this corpus. With one row per event, event grouping imposes the same grouping constraint as random-event assignment, but its deterministic fold allocation differs and therefore produces different scores. It is reported as an intermediate control. The protocol remains reusable on datasets with more multi-event sequences. In the present sample, however, the result does not demonstrate practically important leakage from random splitting.

### 5.4 Baseline performance and calibration

Table 4 and Fig. 4 give per-baseline holdout performance ( $n = 430$ ) with sequence-cluster bootstrap 95% intervals. The no-skill B0 has Brier 0.232 ([0.212, 0.253]); its calibration slope cannot be estimated because it assigns the same probability to every event. The learned baselines have lower Brier (B1 0.198, B2 0.181, B4 0.187, B5 0.181) with overlapping intervals; calibration intercept/slope pairs are finite and positive. No baseline is claimed to win; the table reports proper scores, uncertainty, and derived calibration states only.

### 5.5 Pairwise ranking stability

Table 5 reports all ten pairwise Brier-score comparisons. Every learned baseline performs better than the no-skill reference B0 (all four intervals exclude zero). Among the learned models, the B2–B4 and B2–B5 comparisons are *unresolved* because their intervals include zero. The remaining learned-model comparisons resolve directionally. In the joint bootstrap ordering, B2 has the highest probability of being uniquely best



**Fig. 1** Cohort construction and outcome-independent role assignment on the frozen USGS DYFI  $M \geq 5$  snapshot. The 7,520 source records yield 2,362 eligible events in 1,885 seismic sequences. Of these, 2,360 are assigned to train ( $n = 1,573$ ), development ( $n = 357$ ), or temporal holdout ( $n = 430$ ); two straddle events are excluded from role assignment.

**Table 1** Cohort accounting and outcome-independent role assignment for the frozen USGS DYFI  $M \geq 5$  snapshot.

| Stage                                               | Events | Severe | Severe rate | Sequences |
|-----------------------------------------------------|--------|--------|-------------|-----------|
| USGS DYFI records with $M \geq 5$ (frozen snapshot) | 7520   | –      | –           | –         |
| Eligible (at least 10 felt reports)                 | 2362   | 703    | 0.298       | 1885      |
| Train                                               | 1573   | 449    | 0.285       | –         |
| Development                                         | 357    | 102    | 0.286       | –         |
| Construction (train+dev)                            | 1930   | 551    | 0.285       | 1530      |
| External temporal holdout                           | 430    | 151    | 0.351       | 354       |
| Temporal-straddle excluded                          | 2      | 1      | –           | –         |

*Note.* Roles were fixed without using outcomes. Two source-verification records were removed before the 2,362-event eligible cohort was constructed and are not included in these rows. Severe = 1 when an event’s maximum community decimal intensity (CDI) is at least 6. The eligible severe count reconciles as  $703 = 449$  (train) +  $102$  (development) +  $151$  (holdout) +  $1$  (straddle exclusion). Thus the 2360 role-assigned events contain 702 severe events.

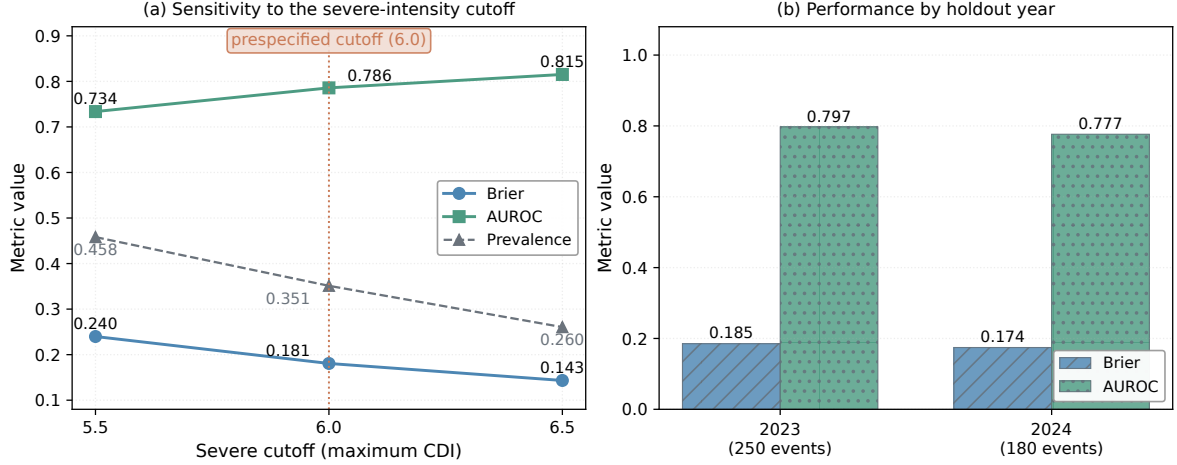
(0.613), followed by B5 (0.387), but the direct B2–B5 comparison remains unresolved. These results therefore do not justify naming a winning model.

The preregistration allowed an “equivalent” verdict but did not specify a positive numerical equivalence margin; its resolution floor was zero. We therefore do not invent a post-hoc margin or label any models equivalent. Instead, we report each point difference, 95% interval, and resolved/unresolved conclusion. Table 6 records

this interpretive restriction, which changes no result.

## 5.6 Transport slices

Table 7 and the right panel of Fig. 2 report the temporal and geographic performance checks for the source-only logistic regression. The two hold-out years are internally consistent (2023:  $n = 250$ , Brier 0.185, AUROC 0.797; 2024:  $n = 180$ , Brier 0.174, AUROC 0.777). The geographic



**Fig. 2** Sensitivity analyses for the source-only logistic regression (B2). (a) Performance as the severe-intensity cutoff varies. (b) Performance within the 2023 ( $n = 250$ ) and 2024 ( $n = 180$ ) portions of the temporal holdout. Lower Brier score and higher AUROC are better.

**Table 2** Sensitivity to the severe-intensity cutoff on the temporal holdout, evaluated with the source-only logistic regression (B2).

| Maximum CDI cutoff | Severe rate | Brier score | AUROC |
|--------------------|-------------|-------------|-------|
| 5.5                | 0.458       | 0.240       | 0.734 |
| 6.0 <sup>a</sup>   | 0.351       | 0.181       | 0.786 |
| 6.5                | 0.260       | 0.143       | 0.815 |

*Note.* <sup>a</sup> Prespecified primary cutoff ( $\geq 6$ ). CDI denotes community decimal intensity. In the construction sample, 0.164 of events lie within 0.5 CDI units of the primary cutoff, and the no-skill Brier score is 0.204. Lower Brier score and higher AUROC indicate better performance.

check spans 82 leave-region-out  $10^\circ$  cells (per-cell Brier 0.002–0.825, median 0.151); these are single-source transport checks, not cross-system generalization claims.

## 5.7 Exclusions, deviations, and reproducibility

Table 6 explains the protocol decisions most relevant to interpretation: excluded comparators and records, the absence of an equivalence claim, and integrity-verification safeguards. Table 8 summarizes the reproducibility safeguards and verification results. The prepared supplement contains the complete SHA-256 fingerprints and machine-readable verification details. All 83 substantive analysis files matched their recorded fingerprints. Three bookkeeping records are self-referential—they cannot contain their own final fingerprints—and are therefore reported separately rather than counted as analysis-file mismatches.

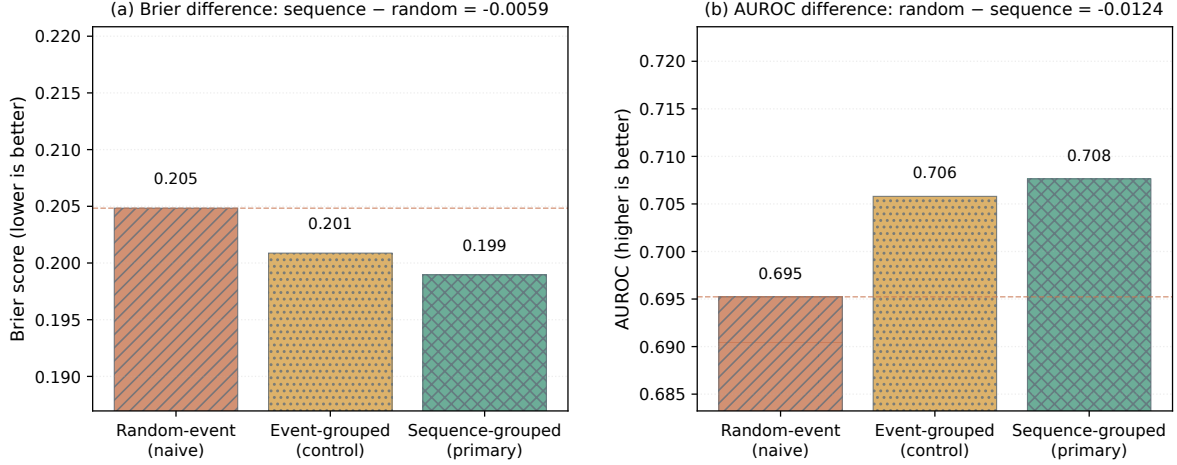
## 6 Discussion

### *Benchmark utility.*

The resource’s utility does not rest on any model outcome. It rests on four reusable properties, each demonstrated here: an executable leakage audit that rejects seeded contamination and accepts only the approved feature set; prespecified outcome-independent roles that a third party can reproduce; calibrated, uncertainty-bearing baselines with a genuine no-skill comparator; and exact reconstruction anchored by recorded digital fingerprints and a single-use evaluation safeguard. A later study can evaluate a new method against this fixed contract without reconstructing a contaminated split, which is the practical benefit the field currently lacks [10, 11, 14].

### *Interpreting the null.*

The observed leakage-inflation point estimate is small and slightly reversed on this corpus. This



**Fig. 3** Five-fold performance of the fixed random-forest diagnostic under random-event, event-grouped, and sequence-grouped fold assignment in the construction sample. The small, slightly reversed differences provide no evidence of performance inflation from random splitting in this sample. Dashed lines mark the corresponding random-event score in each panel.

**Table 3** Cross-validated performance of the fixed random-forest diagnostic (B4) under three fold-assignment schemes in the construction sample.

| Fold assignment            | Brier score | AUROC  | Log-loss | Folds |
|----------------------------|-------------|--------|----------|-------|
| Random-event (naive)       | 0.2048      | 0.6952 | 0.9640   | 5     |
| Event-grouped (control)    | 0.2009      | 0.7058 | 0.9223   | 5     |
| Sequence-grouped (primary) | 0.1990      | 0.7076 | 0.9709   | 5     |

*Note.* Using unrounded values, the sequence-grouped minus random-event Brier difference is -0.0059; the random-event minus sequence-grouped AUROC difference is -0.0124. These are descriptive point estimates; no uncertainty interval was prespecified for this contrast. Because there is one row per event, event grouping imposes the same grouping constraint as random-event assignment, but its deterministic fold allocation is different and can produce different scores.

is a legitimate, honestly reported benchmark finding rather than a defect: on a one-row-per-event  $M \geq 5$  universe of largely singleton sequences the naive random split does not materially inflate optimism, and sequence grouping has little practical effect here. The protocol remains useful for future datasets containing more multi-event sequences, but the present result does not demonstrate leakage. We do not reframe the reversed gap as evidence of leakage or as a win.

#### *No winning model.*

Every learned baseline separates from the no-skill reference. However, the direct B2–B5 comparison between the two lowest-Brier models is unresolved, as is B2–B4, so the results do not establish a decisive winner. Consistent with the resource’s scope, we claim no winning model and no operational, real-time, lead-time, cross-system, or causal benefit.

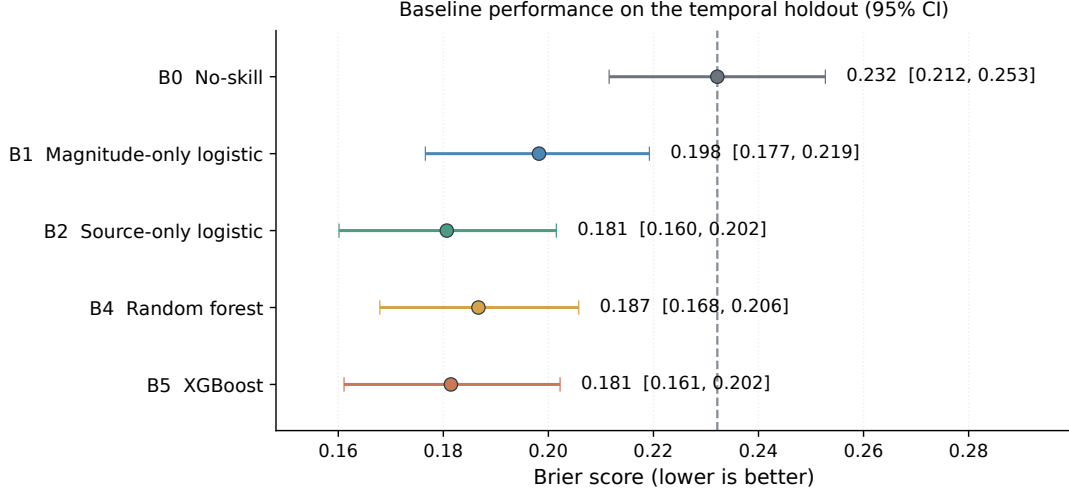
## 7 Limitations and threats to validity

### *Magnitude dominance.*

The severe endpoint is plausibly magnitude-dominated; the magnitude-only baseline B1 is deliberately included as a strong simple comparator, and the thin learned-versus-learned separation is consistent with this. This bounds, and is disclosed as, a limitation on baseline potential improvement rather than a flaw in the protocol.

### *Ascertainment.*

CDI depends on responder count and coverage; the ten-response eligibility filter mitigates but does not remove ascertainment. Response count is used only for eligibility and is excluded from model inputs.



**Fig. 4** Brier scores and 95% sequence-cluster bootstrap confidence intervals for the five baselines on the temporal holdout ( $n = 430$ ). Lower values are better; the vertical dashed line marks the no-skill B0 score. B3 is omitted because it contains event year, which would reveal the temporal holdout period (Table 6).

**Table 4** Per-baseline performance on the temporal holdout ( $n = 430$ ) with sequence-cluster bootstrap 95% CIs and derived calibration states.

| Baseline                     | Brier | 95% CI         | Log-loss | AUROC | Calibration <sup>a</sup> | Accuracy |
|------------------------------|-------|----------------|----------|-------|--------------------------|----------|
| B0 (no-skill)                | 0.232 | [0.212, 0.253] | 0.658    | 0.500 | INFEASIBLE               | 0.65     |
| B1 (magnitude-only logistic) | 0.198 | [0.177, 0.219] | 0.583    | 0.719 | 0.32 / 1.02              | 0.70     |
| B2 (source-only logistic)    | 0.181 | [0.160, 0.202] | 0.536    | 0.786 | 0.64 / 1.29              | 0.73     |
| B4 (random forest)           | 0.187 | [0.168, 0.206] | 0.553    | 0.777 | 0.74 / 1.48              | 0.73     |
| B5 (XGBoost)                 | 0.181 | [0.161, 0.202] | 0.541    | 0.783 | 0.42 / 1.07              | 0.74     |

*Note.* <sup>a</sup> Intercept / slope. These values cannot be estimated for B0 because it assigns the same probability to every event. B3 includes event year and is therefore omitted from the temporal holdout (Table 6). Accuracy uses the probability cutoff of 0.5 fixed on the development set. Lower Brier score and log-loss are better; higher AUROC and accuracy are better. The two lowest-Brier models cannot be distinguished in their direct paired comparison.

### *Single-source scope and transport.*

The resource is single-source (USGS DYFI); the geographic slices are within-system transport checks and do not establish equivalence with another earthquake catalogue. Cross-system transport is deliberately excluded from scope.

### *Limited effect of sequence grouping in this corpus.*

Because most sequences contain only one event, the random-versus-sequence-grouped contrast is close to zero. The empirical consequence in this dataset is therefore modest; the reusable contribution is the evaluation protocol.

### *Uncertainty of the split diagnostic.*

The random-versus-sequence comparison is a descriptive point estimate; no uncertainty interval was prespecified for this contrast. We therefore

make neither an equivalence claim nor an inferential claim that the true difference is zero.

### *Held-out semantics.*

The external holdout is held out from model development with aggregate feasibility previously known; it is not “untouched” and this manuscript does not describe it as untouched or confirmatory.

### *Public release.*

Version 1 of the data, code, frozen results, and reproduction instructions is available in a tagged public repository at <https://github.com/akssha74/dyfi-benchmark-reliability>. The release retains complete digital fingerprints so that later revisions remain distinguishable from the version evaluated here.

**Table 5** Pairwise comparison of baseline Brier scores on the temporal holdout using a paired sequence-cluster bootstrap.

| Model A | Model B | A – B   | 95% CI            | $P(A \text{ better})$ | Conclusion |
|---------|---------|---------|-------------------|-----------------------|------------|
| B0      | B1      | 0.0339  | [0.0171, 0.0511]  | 0.000                 | resolved   |
| B0      | B2      | 0.0515  | [0.0342, 0.0681]  | 0.000                 | resolved   |
| B0      | B4      | 0.0455  | [0.0307, 0.0598]  | 0.000                 | resolved   |
| B0      | B5      | 0.0507  | [0.0319, 0.0682]  | 0.000                 | resolved   |
| B1      | B2      | 0.0176  | [0.0101, 0.0249]  | 0.000                 | resolved   |
| B1      | B4      | 0.0115  | [0.0023, 0.0205]  | 0.009                 | resolved   |
| B1      | B5      | 0.0168  | [0.0055, 0.0273]  | 0.001                 | resolved   |
| B2      | B4      | -0.0060 | [-0.0118, 0.0002] | 0.970                 | unresolved |
| B2      | B5      | -0.0008 | [-0.0078, 0.0068] | 0.613                 | unresolved |
| B4      | B5      | 0.0052  | [0.0003, 0.0099]  | 0.017                 | resolved   |

*Note.* Lower Brier score is better, so a negative A – B difference favors model A. “Resolved” means that the 95% interval excludes zero; “unresolved” means that it includes zero. Because no numerical equivalence margin was specified before analysis, unresolved comparisons are not labelled equivalent. B0 is the no-skill reference; B1 is magnitude-only logistic regression; B2 is source-only logistic regression; B4 is random forest; and B5 is XGBoost.

**Table 6** Excluded analyses and safeguards affecting interpretation.

| Decision                          | Treatment                     | Rationale                                                                                                                                            |
|-----------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| B3 expanded-metadata model        | Omitted from temporal holdout | It includes event year, which would reveal the evaluation period.                                                                                    |
| Published DYFI intensity equation | Omitted as a comparator       | It requires unavailable site distance and tectonic-region inputs, plus an unvalidated conversion to event probabilities.                             |
| Two pre-screened records          | Excluded before analysis      | This prevents information seen during manual verification from entering an analysis role.                                                            |
| Equivalence interpretation        | Descriptive comparisons only  | No numerical equivalence margin was specified before analysis, so the paper reports resolved or unresolved differences without claiming equivalence. |
| Integrity verification            | All 83 analysis files matched | Three self-referential bookkeeping records that cannot contain their own final fingerprints are reported separately.                                 |

*Note.* These decisions were recorded before, or disclosed with, the analysis. No result, subset, cutoff, or comparison was omitted because of its observed outcome.

## 8 Conclusion

We release a versioned, digitally fingerprinted, single-source USGS DYFI benchmark for retrospective severe felt-intensity prediction from outcome-blind source metadata. The benchmark provides a reusable, leakage-audited evaluation contract, reference baselines, and verification tools. On the frozen snapshot, the observed split difference is small and slightly reversed, every learned baseline separates from the no-skill reference, and comparisons among the lowest-Brier models do not establish a winner. Every reported number reproduces exactly from the frozen analysis files. The resource lets subsequent macroseismic-intensity work evaluate against a fixed leakage-safe protocol, and remains reusable

on denser-sequence datasets containing more multi-event sequences.

**Acknowledgements.** Generative AI tools, including ChatGPT, were used for language editing, formatting assistance, and code review. The authors independently reviewed and verified all outputs and take full responsibility for the final text, code, citations, and quantitative results. No AI-generated figures are included. The underlying data are provided by the U.S. Geological Survey “Did You Feel It?” program and the Advanced National Seismic System (ANSS) Comprehensive Catalog.

**Table 7** Temporal and geographic performance checks on the holdout using the source-only logistic regression (B2).

| Subset                | Events | Brier score  | AUROC |
|-----------------------|--------|--------------|-------|
| Temporal 2023         | 250    | 0.185        | 0.797 |
| Temporal 2024         | 180    | 0.174        | 0.777 |
| Geographic (82 cells) | 430    | 0.151 (med.) | –     |

*Note.* The geographic check covers 82 held-out  $10^\circ$  grid cells. Its entry is the median cell-level Brier score; cell-level values range from 0.002 to 0.825 and are included in the prepared supplementary results. These are within-USGS checks and do not establish transfer to another data system.

**Table 8** Reproducibility safeguards and verification results.

| Check                      | Recorded result                                   | What it establishes                                                   |
|----------------------------|---------------------------------------------------|-----------------------------------------------------------------------|
| Frozen source data         | 7,520 USGS DYFI records                           | A complete SHA-256 digital fingerprint is archived.                   |
| Protected holdout use      | Opened once                                       | A single-use safeguard prevented repeat evaluation.                   |
| Independent reconstruction | Two clean reruns                                  | Both reproduced the recorded results and holdout predictions exactly. |
| Model fitting              | 91 fits (prespecified cap: 120)                   | The recorded run remained within its computational budget.            |
| Software environment       | Python 3.11.15; package versions archived         | Exact versions are fixed for reproduction.                            |
| Integrity audit            | 83 of 83 analysis files matched                   | Three self-referential bookkeeping records are reported separately.   |
| Licensing                  | CC0-1.0 data; MIT code; USGS public-domain source | Terms permit independent reuse and verification.                      |

*Note.* Complete digital fingerprints, environment details, and machine-readable verification records are included in the tagged public release.

## Declarations

- **Funding.** No specific grant from funding agencies in the public, commercial, or not-for-profit sectors was received for this work.
- **Competing interests.** The authors declare that they have no competing interests.
- **Ethics approval and consent to participate.** Not applicable; this study involves no human participants, no new human-subjects collection, and no animals. It reuses aggregate, public-domain, event-level metadata; no respondent-level personally identifying information is accessed or redistributed.
- **Consent for publication.** Not applicable.
- **Data availability.** The version 1 derived event-level benchmark, frozen results, manifests, and reproduction instructions are publicly available under CC0 1.0 at <https://github.com/akssha74/dyfi-benchmark-reliability/releases/tag/v1.0.2>. The underlying source data are USGS “Did You Feel It?” and the ANSS Comprehensive Catalog (public domain; [10.5066/F7J101C8](https://doi.org/10.5066/F7J101C8) and [10.5066/F7MS3QZH](https://doi.org/10.5066/F7MS3QZH)) and are reconstructable via the released fetch-and-verify source-locator manifest. No respondent-level personally identifying information is redistributed.
- **Code availability.** The resource code (leakage audit, sequence grouping, role assignment, baselines, metrics, protected-evaluation runner, and a dependency-free verifier that recomputes every reported number from the frozen per-event outputs) is released under the MIT licence in the same tagged public repository as the data.
- **Author contributions.** A. Sharma designed the benchmark protocol and instrumentation, implemented the analysis, and drafted the manuscript. L. Prasad supervised the work and revised the manuscript. Both authors read and approved the final manuscript.

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